

Feedback Control Systems (EE3004)**Sessional-II Exam**Date: April 10th 2025

Total Time (Hrs): 1

Course Instructor

Total Marks: 30

Dr. Omer Saleem & Ms. Tamania Javed

Total Questions: 3

Roll No

Section

Student Signature

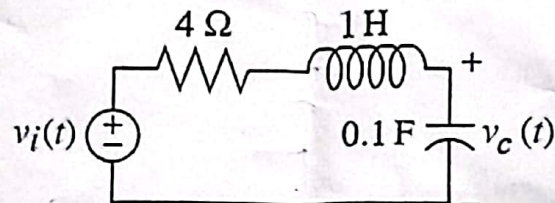
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Attempt all the questions.

CLO # 2: Analyze the transient and steady state response of a feedback system.Q1: Referring to the following 2nd order electrical system,

[10 marks]

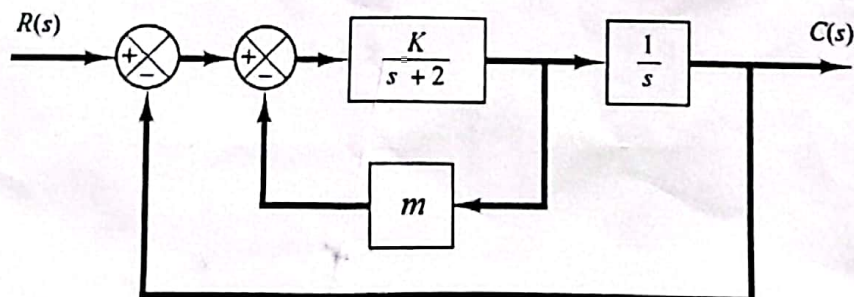
- Find the system's transfer function, $G(s) = v_c(s) / v_i(s)$.
- Calculate the peak time (T_p), settling time (T_s), and overshoot (%OS) of the system's response.

**CLO # 2: Analyze the transient and steady state response of a feedback system.**

Q2: Referring to the following system,

[10 marks]

- Find the system's overall transfer function, $T(s)$, by reducing the block diagram.
- Calculate the values of K and m such that the system has a damping-ratio $\zeta = 0.8$ and a natural-frequency $\omega_n = 4.0$ rad/sec.



CLO # 3: Analyze the stability of a system.

Q3: Referring to the following closed loop transfer function of a system,

[10 marks]

$$T(s) = \frac{s + 8}{s^5 - s^4 + 4s^3 - 4s^2 + 3s - 2}$$

- a. Find the closed loop system's number of poles in the right half-plane, the left half-plane, and on the $j\omega$ -axis.
- b. Analyze whether the system is stable or not.